



Jyothi Engineering College(Autonomous)

DEPARTMENT OF MECHATRONICS ENGINEERING

Vth Semester - B.Tech

First Internal Examination, Aug 2026

13-Aug-2026

PEMRT525 - IOT AND APPLICATIONS

Total Mark: 30

CODE: MRU5X130826F

Total Time: 1 Hrs : 15 Mins

CO1	Apply IoT fundamentals to model and explain IoT architectural views	Applying(P)
CO2	Apply knowledge of smart objects, sensors, actuators, and functional blocks in IoT systems	Applying(P)
		CO BL MARK

PART A

1. Define IoT. List and recall any two characteristics of IoT with a one-line description of each. CO1 1 (3)
2. Describe the four layers of IoT system architecture and explain the role of each layer in the data flow. CO1 2 (3)
3. Define M2M. State the role of an M2M gateway using an example to illustrate how it connects an M2M area network to an external network. CO2 1 (3)
4. Classify sensors using any three criteria. Provide one sensor example and one IoT application for each criterion. CO2 2 (3)

PART B

5. Apply knowledge of the Physical Design of IoT to describe the four-layer IoT protocol stack. Use relevant examples to justify the selection of protocols at each layer for IoT environments. CO1 3 (9)

OR

6. Apply your understanding of IoT communication models to classify and describe all four models with examples. Demonstrate the differences by summarising them in a comparison table. CO1 3 (9)
7. Use a labelled diagram to illustrate the architecture of an M2M system. Apply your understanding to compare IoT and M2M systems using a table with at least five points of difference. CO2 3 (9)

OR

8. Illustrate WSN architecture with a labelled diagram. Apply the FFD and RFD classification to describe node capabilities and their differences. Use knowledge of the six design constraints of WSN nodes to explain how data aggregation and self-organisation address these constraints. CO2 2 (9)